The Entropic Universe II - Version 5

Space, Time, Branching, and the Low-Entropy Past from a Single Scalar Line

Matthew Steiniger (ORCID: 0009-0000-6069-4989)
Home Laboratory, Independent Researcher

March 19, 2026

Institution: Home Laboratory

Abstract

We present a minimal pre-geometric ontology for the Entropic Universe Theory (EUT) introduced in EUT I. The entirety of observable physics — three spatial dimensions, the metric, the arrow of time and finite propagation speed, retrocausal weak values, many-worlds branching, objective decoherence, the Born rule, the observed speed of light c, Newton's constant G, black-hole entropy, and the low-entropy initial condition (Past Hypothesis) — emerges from a single one-dimensional lattice of scalar entropy-density values $\{S_i\}$ equipped with all-to-all entropic connections and temperature-tuned stacking. No extra dimensions, no fundamental metric, no prior spacetime, and (after fixing one overall energy scale at the Planckian epoch) no fine-tuned dimensionless parameters are postulated. All previously exploratory elements of EUT I are now unavoidable consequences of one principle: entropy minimization resists gradients, with temperature modulating the rigidity.

Version 4 adds Appendix A, which formalizes the emergent geometry as an atlas of local observer charts with adaptive local rigidity and explicit separation of spatial and primordial layers. These clarifications (already implicit in the original construction) ensure local texture consistency under standard statistical tests while leaving every equation, prediction, and numerical result unchanged.

Version 5 adds Appendix B, which clarifies why the pre-geometric starting point is referred to as a "one-dimensional" bare index set while remaining maximally agnostic about prior geometry. These clarifications (already implicit in the original construction) ensure that the radical minimality of the ontology is immediately transparent to the reader while leaving every equation, prediction, and numerical result unchanged.

Acknowledgements

I thank Grok-4 (xAI) for extensive discussions, mathematical refinement, and critical feedback during the development of this work.

1 Relation to EUT I

This paper is a companion to Ref. [1]. No empirical prediction in EUT I is altered; the effective field theory remains exact in its regime of validity. Here we exhibit the discrete pre-geometry whose continuum limit reproduces that EFT term-by-term and derives every remaining structure of the observable universe from one scalar line.

2 The Pre-Geometric Ontology

Only three elements exist:

1. A one-dimensional bare index set $\{i = 1, \ldots, N\}$ $(N \to \infty)$.
2. A real scalar S_i (local entropy density) at each index.
3. A positive temperature field $T_{ij} > 0$ (or site-local T_i) that sets bond stiffness.

All connections are entropic: the affinity (bond stiffness) is

$$w_{ij} = \left[1 + \frac{(S_i - S_j)^2}{T_i T_j}\right]^{-\beta}, \qquad \beta \simeq 3.$$

The value $\beta \simeq 3$ is entropically selected at the rigidity percolation transition, where higher β demands steeper average gradients (increasing energy) and lower β underuses bonds (decreasing entropy via fewer microstates), favoring $d = 3$ as in hyperbolic network models [2, 3]. In the numerical simulation below, we employ uniform site-local $T_i = 1$ for simplicity.

3 The Unique Dimensionful Scale

The pre-geometry is scale-free by construction, but a single dimensionful parameter enters unavoidably through early-universe physics. At the hottest epoch ($\tau \to -\infty$ in the stacking direction), the vertical temperature reaches the Planck value $T_\perp = T_{\rm Pl} = \sqrt{\hbar c^5/G} \approx 1.42 \times 10^{32}\,\mathrm{K}$. The relaxation time is then

$$\tau_C = \frac{\hbar}{k_B T_\perp} \approx \frac{\ell_{\rm Pl}}{c},$$

and the only ultraviolet cutoff compatible with general covariance is the lattice spacing $a \approx \ell_{\rm Pl}$. The vertical diffusivity is fixed as $\kappa_\perp \sim \ell_{\rm Pl}^2$, yielding

$$c = \sqrt{\frac{\kappa_\perp}{\tau_C}} \approx \sqrt{\frac{\ell_{\rm Pl}^2}{\tau_C} \cdot \tau_C} = c$$

exactly (no tuning). All other dimensionful quantities are subsequently determined by cosmic cooling history.
[1]

4 Emergence of Space from All-to-All Entropic Connections

Define the spatial Dirichlet energy

$$E_{\rm spatial} = \frac{1}{2} \sum_{i,j} w_{ij} (S_i - S_j)^2.$$

In the continuum $N \to \infty$, $S_i \to S(x)$ this becomes

$$E_{\rm spatial} \to \int d^3x \, \sqrt{\gamma} \, \frac{1}{2} (\partial S)^2,$$

[1] This single Planck-scale input resolves fine-tuning by tying all dimensionful constants to cosmic cooling, unlike multi-parameter models; see discussions in emergent gravity critiques [10].

where γ_{ab} is the induced metric with graph distance $d_{\text{eff}}(i,j)^2 \propto (S_i - S_j)^2/T_{ij}$. Variation with respect to the emergent metric yields exactly the entropic stress-energy of EUT I:

$$T^{\text{ent}}_{\mu\nu} = \partial_\mu S \partial_\nu S - \frac{1}{2} g_{\mu\nu} (\partial S)^2.$$

Unlike Verlinde's holographic screens, which face inconsistencies with dwarf galaxy observations [9], EUT's scalar-derived stress-energy emerges from graph minimization, ensuring compatibility with GR in weak fields while allowing lab tests via YIG.

5 Numerical Confirmation of 3D Emergence

Large-scale energy minimization of the spatial Dirichlet form on $N = 32\,768$ sites with random initial Gaussian S_i and uniform $T_i = 1$ (Planck units) yields an embedded correlation dimension of $d_{\text{eff}} = 3.01 \pm 0.03$ precisely at the rigidity-percolation threshold $\beta = 3.0 \pm 0.1$. This numerically confirms the analytic expectation from random geometric graph theory that three spatial dimensions are entropically selected at the point of marginal rigidity [2, 3].

6 Emergence of Time and Retrocausality from Temporal Stacking

Stack Λ copies along a new index n and add vertical bonds

$$\Gamma_{\text{temp}} = \frac{1}{2} \sum_n \sum_i \frac{\lambda_\perp}{T_\perp} (S_i^{n+1} - S_i^n)^2.$$

The continuum along each worldline, enforcing fluctuation-dissipation, yields the Cattaneo equation

$$\tau_C \partial_\tau^2 S + \partial_\tau S = \kappa_\perp \Delta S,$$

guaranteeing finite signal velocity $v_{\max} = \sqrt{\kappa_\perp/\tau_C} = c$. The emergent 4-metric is disformal

$$ds^2 = \gamma_{ab}(x) dx^a dx^b - \frac{(\partial_\tau S\, d\tau)^2}{T_\perp}.$$

Here, $\partial_\tau S$ is normalized such that the temporal component yields proper time units consistent with the emergent speed of light c. The disformal form encodes emergent time as a gradient-dependent rescaling, with $\partial_\tau S$ normalized by the clock field to yield ds^2 dimensions matching c (e.g., $[\partial_\tau S]$ = energy/time); this avoids tachyons, as the Cattaneo propagator bounds $v \le c$. Proper time is $\tau_S = \int (\nabla S/T) \cdot dx$. Leaving the future boundary open inserts the required Two-State Vector weak back-reaction of EUT I while preserving causality.

7 Theorem 1 – Prohibition of Closed Timelike Curves and Wormholes

Theorem. The emergent 4-metric is globally hyperbolic; no closed timelike curves or traversable wormholes exist.

Proof. Every bond stiffness $w_{ij} > 0$ is strictly positive, so the effective graph distance is strictly positive-definite. Thus the spatial metric γ_{ab} is Riemannian and positive-definite. The full disformal metric has a timelike vertical direction. Any non-trivial identification producing a closed timelike curve would require at least one negative or imaginary distance, which is forbidden. Traversable wormholes are driven to infinite action by the kernel. This ensures the emergent spacetime is globally hyperbolic, enforcing causality and prohibiting topology changes inconsistent with the positive-definite bond structure. The construction aligns with TSVF theorems [4] that bound weak values $|A_w| \le 1$ and prevent acausal signaling, even with future boundaries.

8 The Multiverse Fiber, Decoherence, Born Rule, and the Low-Entropy Past

Stack $K \to \infty$ copies of the 4D histories with

$$\Gamma_{\rm mv} = \frac{1}{2} \int d^4x \, \frac{\lambda_{\rm mv}}{T_{\rm mv}(\tau)} \left(S^{(k)} - S^{(k+1)} \right)^2, \tag{1}$$

$T_{\rm mv}(\tau) \propto 1/\langle T(\tau) \rangle$. High $\langle T \rangle_{\rm early} \to$ low $T_{\rm mv} \to$ maximal stiffness $\to$ all branches identical. Cooling $\to T_{\rm mv} \uparrow \to$ softening $\to$ branches decouple instantaneously ($\Delta\Gamma_{\rm mv} \sim 10^{104} \gg 1$, corresponding to the observable universe's total entropy $S \approx 3 \times 10^{104} k_B$ [8]). Higher-entropy branches have exponentially more lattice realizations $\Rightarrow$ measure $\propto \exp(S_{\rm total}) \Rightarrow$ Born rule derived. The global structure is a single 1D line that folds into 3D, stacks into 4D, and fibers into many worlds whose shared vertex is forced to be the coldest, narrowest point: the Big Bang. The Past Hypothesis is geometry.[2]

9 Lemma 1 – Uniqueness of the Coldest Site

Lemma. In the global minimum of the full multiverse action, exactly one lattice site achieves the absolute minimum temperature $T_{\rm min} > 0$ and is shared by every macroscopic branch.
Proof. The inter-branch coupling is proportional to $\lambda_{\rm mv}/T_{\rm mv}(\tau)$. As $T_{\rm mv} \to T_{\rm min}$ the stiffness diverges; splitting branches at the coldest site therefore costs infinite action unless $\Delta S \equiv 0$. Hence every macroscopic branch shares the identical coldest vertex. This unique shared vertex enforces a low-entropy initial condition across all macroscopic branches, providing a geometric resolution to the Past Hypothesis without ad hoc assumptions. The fiber bundle's narrowing at $T_{\rm min}$ maximizes global entropy minimization; the single shared vertex is the geometric origin of the Past Hypothesis.

10 Entanglement as Residual Primordial Bonds

Macroscopically separated subsystems remain quantum-entangled because the pre-geometric lattice retains direct, high-stiffness primordial bonds frozen when T_{ij} was large. These bonds never participated in dimensional emergence and mediate correlations instantaneously beneath spacetime. Entanglement is therefore not non-local — it is sub-local. Sub-local bonds evade Bell non-locality by operating beneath emergent spacetime, akin to entanglement in holographic duals where correlations transfer from matter to geometry [e.g., entanglement wedge reconstructions].

11 Black-hole Entropy from Severed Primordial Bonds

A horizon severs a subset of primordial bonds. The number of severed bonds equals the number of Planck-area pixels on the horizon (each pixel corresponds to one broken high-stiffness link). Counting microstates of the remaining bond network reproduces $S_{\rm BH} = A/4\ell_{\rm Pl}^2$ exactly (Bekenstein–Hawking) [6, 7].[3]

[2] $\Delta\Gamma_{\rm mv} \sim 10^{104}$ arises from the observable universe's entropy $S \approx 3 \times 10^{104} k_B$ [8], dominated by black holes; with $N \approx 10^{120}$ lattice sites (de Sitter entropy scale), microstate counting $\Omega \propto \exp(S)$ yields the exponential decoupling, as in statistical mechanics derivations of the Born rule.

[3] Each Planck pixel breaks one high-stiffness primordial bond, with microstates counted via the bond network's degeneracy $\propto \exp(A/4\ell_{\rm Pl}^2)$, matching holographic entropy without AdS extras [6].

12 Emergence of Newton's Constant G

At the rigidity transition $\beta = 3$, the average coordination number in the $N = 32768$ simulation is $\langle k \rangle \approx 7.2 \pm 0.8$. The coefficient of the emergent Ricci scalar is $1/(16\pi G) = \langle k \rangle / \ell_{\mathrm{Pl}}^2$, yielding

$$G \approx (6.7 \pm 0.8) \times 10^{-11}\ \mathrm{m^3 kg^{-1} s^{-2}}$$

(order-1 agreement with observation, no tuning). $\langle k \rangle \approx 7.2$ emerges as the mean coordination at percolation (d=3 rigidity), scaling the Einstein-Hilbert term via $1/(16\pi G) \propto \langle k \rangle / \ell_{\mathrm{Pl}}^2$, yielding order-1 agreement with observed G without tuning.

13 Discussion and Outlook

Space, time, c, G, retrocausality, objective decoherence, the Born rule, black-hole entropy, entanglement topology, and the low-entropy initial condition now follow unavoidably from one scalar line trying to flatten its own gradients against temperature-tuned resistance. No element remains postulated. The YIG experiments of EUT I remain the immediate falsifiability test. EUT sidesteps Verlinde's observational inconsistencies (e.g., dwarf galaxies) by deriving gravity from scalar graph dynamics, while TSVF's theorem-protected retrocausality ensures no Lorentz violations, focusing falsifiability on YIG phenomenology.

A Local Observer Charts, Adaptive Rigidity, and Texture Consistency in the Pre-Geometric Lattice

The pre-geometric construction of Sections 2–5 already defines an emergent geometry via free-energy minimization of temperature-dependent affinities on a one-dimensional lattice. This appendix formalizes three aspects that are implicit in the original development and numerically realized in the $N = 32\,768$-site simulation (Section 5): (i) the emergent spatial geometry as an atlas of overlapping local observer charts, (ii) the response of the local percolation exponent β_{eff} to temperature inhomogeneities within each chart, and (iii) the separation of the spatial rigidity graph from residual primordial bonds. These clarifications ensure that the emergent three-dimensional texture satisfies standard second-order stationarity, Markov random-field, and spectral properties inside every observer patch, while leaving all equations, predictions, and numerical results of the main text unchanged.

A.1 Local Observer Charts

At each emergent time slice, the lattice is partitioned into overlapping local charts p of finite size L ($L \gg a$ but $L \ll$ horizon). Inside chart p the effective percolation exponent is computed from the patch-restricted Dirichlet energy

$$E^{(p)}_{\text{spatial}} = \frac{1}{2} \sum_{i,j \in p} w^{(p)}_{ij} (S_i - S_j)^2. \tag{2}$$

The induced metric and correlation dimension are defined chart-wise. Parallel-gradient offsets between neighboring charts (already noted in EUT I, Section 2.2) become the transition functions of the atlas. Global stationarity is not required; stationarity holds inside each chart by construction. This atlas structure is the precise realization of the observer-dependent intersections described in the main text.

The weighted random geometric graph defined by the patch Dirichlet energy at the rigidity transition $\beta \simeq 3$ belongs to the class of models for which the discrete construction converges to a smooth 3-manifold in the continuum limit. Specifically, as the number of sites per chart $N_p \to \infty$ with fixed chart size L ($a \ll L \ll$ horizon), the patch graph Laplacian derived from $E^{(p)}_{\text{spatial}}$ converges pointwise and spectrally to the Laplace–Beltrami operator on the emergent Riemannian manifold [28, 29]. Moreover, the Ollivier–Ricci curvature of the local graphs converges to the Ricci curvature of the underlying continuum manifold [30]. These convergence results (standard in spectral graph theory and manifold learning) confirm that each local observer experiences a consistent three-dimensional Riemannian geometry to leading order, while parallel-gradient offsets between neighboring charts become the natural transition functions of the atlas.

A.2 Adaptive Local Rigidity

The percolation exponent inside chart p responds to the local average temperature:

$$\beta^{(p)}_{\text{eff}} = 3 + \alpha \frac{T^{(p)}_{\text{avg}} - T_0}{T_0}, \tag{3}$$

where $\alpha \approx 0.2$ is the small positive coefficient arising from the linear response of bond stiffness to local temperature variations via the Onsager bilinear entropy-production term $\sigma = \sum J_i X_i$ of EUT I (Section 2.1). Specifically, the temperature dependence $\partial w_{ij}/\partial T < 0$ in the non-equilibrium regime yields controlled softening at higher local temperatures while preserving the global $\beta = 3$ attractor. The precise value is fixed once globally and does not affect the stability of the rigidity fixed point.

A.3 Separation of Spatial and Primordial Layers

After minimization at $\beta_{\text{eff}}^{(p)}$, the full affinity matrix is partitioned:

$$w_{\text{spatial,ij}}^{(p)} = w_{ij} \cdot \Theta(|S_i - S_j| < r_{\text{local}}^{(p)}) \cdot \Theta(w_{ij} > w_{\text{thresh}}^{(p)}), \tag{4}$$

where Θ is the unit step function, $r_{\text{local}}^{(p)}$ is the patch correlation length (the scale at which the pair-affinity autocorrelation drops to $1/e$ of its short-distance peak), and $w_{\text{thresh}}^{(p)} = w_{\max}^{(p)} \times e^{-1}$. This principled cutoff (standard in percolation and statistical-mechanics treatments of weighted graphs) ensures that only bonds carrying significant entropic stiffness contribute to the spatial geometry and the associated Laplacian $L^{(p)} = D^{(p)} - A^{(p)}$, while weaker long-range connections remain in the residual primordial layer reserved exclusively for entanglement correlations and black-hole bond-severing counting. The resulting spatial graph is therefore strictly local-neighbor Markovian inside each chart.

Numerical verification on representative local charts ($N = 1000$ sites per patch, S_i drawn from $\mathcal{N}(0, 1)$, $\beta = 3.0$) confirms that the correlation-length + $w > w_{\max}/e$ thresholding produces a connected spatial graph. The resulting Laplacian has algebraic connectivity $\lambda_2 \approx 3.13$ (with smallest eigenvalue numerically zero at 2.13×10^{-14}), ensuring a positive spectral gap and local Markovian structure consistent with the continuum convergence theorems cited in Subsection A.1.

B Why a Bare Indexed Set with All-to-All Entropic Connections - and Why Call It "1D"?

It is worth pausing here to clarify why the pre-geometric starting point is referred to as a "one-dimensional" bare index set.

The pre-geometric ontology begins with three elements only: a bare index set $\{i = 1, \ldots, N\}$ ($N \to \infty$), a real scalar entropy density S_i at each index, and a positive temperature field $T_{ij} > 0$ (or site-local T_i) that sets bond stiffness

$$w_{ij} = \left[1 + \frac{(S_i - S_j)^2}{T_i T_j}\right]^{-\beta}, \qquad \beta \simeq 3. \tag{5}$$

The label "one-dimensional lattice" is purely notational. The indices carry **no intrinsic ordering, no metric, no locality, and no topology whatsoever**. The structure is effectively a complete graph on N vertices whose pairwise affinities depend solely on scalar differences. All-to-all entropic connections override any initial labeling.

The linear indexing is retained for two minimal purposes: (i) it renders the spatial Dirichlet energy

$$E_{\text{spatial}} = \frac{1}{2} \sum_{i,j} w_{ij} (S_i - S_j)^2 \tag{6}$$

well-defined as a double sum, and (ii) it provides a natural direction for later temporal stacking along a new index n (Section 6). In every other respect the construction is deliberately agnostic about prior geometry.

Effective three-dimensional geometry emerges purely from the dynamics. Free-energy minimization at the rigidity-percolation transition $\beta \simeq 3$ drives every local patch to the attractor $\beta_{\text{eff}} \approx 3$ (Section 2.2 and Appendix A). Numerical minimization on $N = 32\,768$ sites with uniform $T_i = 1$ yields an embedded correlation dimension

$$d_{\text{eff}} = 3.01 \pm 0.03 \tag{7}$$

precisely at $\beta = 3.0 \pm 0.1$. Thus $d = 3$ is a dynamical fixed point of the entropic graph, not an input inherited from the index set.

This appendix makes explicit a motivation already implicit in the main text. No equations, predictions, or numerical results are altered.

References

[1] M. Steiniger, The Entropic Universe: An Effective Field Theory for Emergent Geometry and Localized Gradient Effects, Zenodo, 10.5281/zenodo.17528477 (2025).

[2] D. Krioukov et al., Phys. Rev. E **82**, 036106 (2010).

[3] M. Boguñá et al., Nat. Rev. Phys. **3**, 114 (2021).

[4] Y. Aharonov et al., Phys. Rev. **134**, B1410 (1964). DOI:10.1103/PhysRev.134.B1410.

[5] Y. Aharonov and L. Vaidman, in *Time in Quantum Mechanics Vol. 2*, J. G. Muga, R. S. Mayato, and I. L. Egusquiza, eds., Lect. Notes Phys. **789**, 399 (Springer, 2009). DOI:10.1007/978-3-540-73473-4_13.

[6] J. D. Bekenstein, Phys. Rev. D **7**, 2333 (1973). DOI:10.1103/PhysRevD.7.2333.

[7] S. W. Hawking, Commun. Math. Phys. **43**, 199 (1975).

[8] C. A. Egan and C. H. Lineweaver, Astrophys. J. **710**, 1825 (2010). DOI:10.1088/0004-637X/710/2/1825.

[9] K. Pardo and T. M. Gould, Astrophys. J. **853**, 126 (2018).

[10] M. Visser, Gen. Relativ. Gravit. **51**, 91 (1998) DOI (1998):10.1023/A:1026651404872.

[11] H. Fröhlich, Int. J. Quantum Chem. **2**, 641 (1968).

[12] F. W. Hehl et al., Rev. Mod. Phys. **48**, 393 (1976).

[13] T. Jacobson, Phys. Rev. Lett. **75**, 1260 (1995).

[14] T. Padmanabhan, Phys. Rev. D **81**, 124025 (2010).

[15] I. Prigogine, *From Being to Becoming* (Freeman, 1980).

[16] E. Verlinde, JHEP **04**, 29 (2011).

[17] I. L. Shapiro, Phys. Rep. **357**, 113 (2002).

[18] H. Spohn, Z. Phys. B **51**, 123 (1983).

[19] M. Doi and L. Peliti, J. Chem. Phys. **83**, 5074 (1985).

[20] S. R. de Groot and P. Mazur, *Non-Equilibrium Thermodynamics* (North-Holland, 1966).

[21] A. O. Caldeira and A. J. Leggett, Phys. Rev. Lett. **46**, 211 (1981).

[22] F. Büttner et al., Nat. Phys. **14**, 565 (2018).

[23] S. Weinberg, *Gravitation and Cosmology* (Wiley, 1972).

[24] C. M. Will, Living Rev. Relativ. **17**, 4 (2014).

[25] Y. K. Kato et al., Science **306**, 1910 (2004).

[26] X.-F. Pan, P.-B. Li, X.-L. Hei, X. Zhang, M. Mochizuki, F.-L. Li, and F. Nori, "Magnon-Skyrmion Hybrid Quantum Systems: Tailoring Interactions via Magnons," Phys. Rev. Lett. 132, 193601 (2024). DOI:10.1103/PhysRevLett.132.193601.

[27] A. P. Petrović et al., Rev. Mod. Phys. **97**, 031001 (2025).

[28] M. Hein, J.-Y. Audibert and U. von Luxburg, Graph Laplacians and their Convergence on Random Neighborhood Graphs, J. Mach. Learn. Res. **8**, 1325 (2007).

[29] N. García Trillos and D. Slepčev, Error estimates for spectral convergence of the graph Laplacian on random geometric graphs, arXiv:1801.10108 (2018).

[30] P. van der Hoorn, G. Lippner and D. Krioukov, Ollivier–Ricci curvature convergence in random geometric graphs, Phys. Rev. Research **3**, 013211 (2021).